

CELEBAL INTERNSHIP ASSIGNMENT-2

SQL Analysis

NAME : - Anmol Bhatia

DATASET :- Superstore

Objective

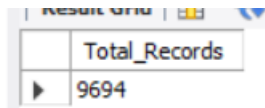
The objective of this assignment is to perform data analysis on the Superstore dataset using SQL by applying filtering, sorting, aggregation, grouping, business use cases, and data validation techniques.

create and use database

```
CREATE DATABASE celebal_assignment2;  
USE celebal_assignment2;
```

count of records

```
SELECT COUNT(*) AS Total_Records  
FROM superstore;
```



Total_Records
9694

first 10 rows of the table

```
SELECT * FROM superstore  
LIMIT 10;
```

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category
1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture
2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture
3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies
4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture
5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10000760	Office Supplies
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10001487	Furniture
7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10002833	Office Supplies
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002275	Technology
9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-10003910	Office Supplies
10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10002892	Office Supplies

Sub-Category	Product Name	Sales	Quantity	Discount	Profit
Bookcases	Bush Somerset Collection Bookcase	261.96	2	0	41.9136
Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs,...	731.94	3	0	219.582
Labels	Self-Adhesive Address Labels for Typewriters b...	14.62	2	0	6.8714
Tables	Bretford CR4500 Series Slim Rectangular Table	957.5775	5	0.45	-383.031
Storage	Eldon Fold 'N Roll Cart System	22.368	2	0.2	2.5164
Furnishings	Eldon Expressions Wood and Plastic Desk Acces...	48.86	7	0	14.1694
Art	Newell 322	7.28	4	0	1.9656
Phones	Mitel 5320 IP Phone VoIP phone	907.152	6	0.2	90.7152
Binders	DXL Angle-View Binders with Locking Rings by S...	18.504	3	0.2	5.7825
Appliances	Belkin F5C206VTEL 6 Outlet Surge	114.9	5	0	34.47

table structure

DESCRIBE superstore;

Result Grid						
Filter Rows:		Export:		Wrap Cell Co		
	Field	Type	Null	Key	Default	Extra
►	Row ID	int	YES		NULL	
	Order ID	text	YES		NULL	
	Order Date	text	YES		NULL	
	Ship Date	text	YES		NULL	
	Ship Mode	text	YES		NULL	
	Customer ID	text	YES		NULL	
	Customer Name	text	YES		NULL	
	Segment	text	YES		NULL	
	Country	text	YES		NULL	
	City	text	YES		NULL	
	State	text	YES		NULL	
	Postal Code	int	YES		NULL	
	Region	text	YES		NULL	
	Product ID	text	YES		NULL	
	Category	text	YES		NULL	
	Sub-Category	text	YES		NULL	
	Product Name	text	YES		NULL	
	Sales	double	YES		NULL	
	Quantity	int	YES		NULL	
	Discount	double	YES		NULL	
	Profit	double	YES		NULL	

products with sales greater than 1000 sorted by sales

SELECT * FROM superstore

WHERE Sales > 1000

ORDER BY Sales ASC;

Result Grid														
Filter Rows:					Export: Wrap Cell Content: 11									
Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit	
Corporate	United States	Oakland	California	94601	West	OFF-ST-10003306	Office Supplies	Storage	Letter Size Cart	1000.02	7	0	290.0058	
Corporate	United States	Seattle	Washington	98103	West	OFF-ST-10003306	Office Supplies	Storage	Letter Size Cart	1000.02	7	0	290.0058	
Home Office	United States	Chicago	Illinois	60653	Central	TEC-PH-10002584	Technology	Phones	Samsung Galaxy S4	1001.584	2	0.2	125.198	
Consumer	United States	Seattle	Washington	98103	West	TEC-PH-10002584	Technology	Phones	Samsung Galaxy S4	1001.584	2	0.2	125.198	
Consumer	United States	Seattle	Washington	98105	West	OFF-ST-10001097	Office Supplies	Storage	Office Impressions Heavy Duty Welded Shelvin...	1003.62	6	0	0	
Consumer	United States	San Francisco	California	94110	West	FUR-TA-10002041	Furniture	Tables	Bevis Round Conference Table Top, X-Base	1004.024	7	0.2	-112.9527	
Home Office	United States	Los Angeles	California	90004	West	FUR-TA-10003715	Furniture	Tables	Hon 2111 Invitation Series Corner Table	1004.976	6	0.2	-175.8708	
Consumer	United States	Cleveland	Ohio	44105	East	OFF-ST-10002011	Office Supplies	Storage	Smead Adjustable Mobile File Trolley with Locka...	1006.056	3	0.2	88.0299	
Home Office	United States	Philadelphia	Pennsylvania	19120	East	FUR-FU-10000576	Furniture	Furnishings	Luxo Professional Fluorescent Magnifier Lamp w...	1007.232	6	0.2	75.5424	
Consumer	United States	Seattle	Washington	98103	West	TEC-PH-10000895	Technology	Phones	Polycom VVX 310 VoIP phone	1007.944	7	0.2	75.5958	
Home Office	United States	Chicago	Illinois	60623	Central	TEC-MA-10002937	Technology	Machines	Canon Color ImageCLASS MF8580Cdw Wireless...	1007.979	3	0.3	43.1991	
Consumer	United States	Charlotte	North Carolina	28205	South	TEC-AC-10002049	Technology	Accessories	Plantronics Savi W720 Multi-Device Wireless He...	1012.68	3	0.2	303.804	
Consumer	United States	San Diego	California	92037	West	FUR-CH-10000785	Furniture	Chairs	Global Ergonomic Managers Chair	1013.488	7	0.2	76.0116	
Corporate	United States	Hialeah	Florida	33012	South	FUR-CH-10000988	Furniture	Chairs	Hon Olson Stacker Stools	1013.832	9	0.2	101.3832	
Consumer	United States	Los Angeles	California	90049	West	OFF-BI-10003527	Office Supplies	Binders	Fellowes PB500 Electric Punch Plastic Comb Bind...	1016.792	1	0.2	381.297	

products from West region

```
SELECT * FROM superstore
WHERE Region = 'West';
```

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	S
1614	CA-2017-138380	12/21/2017	12/25/2017	Standard Class	YS-21880	Yana Sorensen	Corporate	United States	Oakland	California	94601	West	OFF-ST-10003306	Office Supplies	St
6211	CA-2017-119809	8/18/2017	8/25/2017	Standard Class	YS-21880	Yana Sorensen	Corporate	United States	Seattle	Washington	98103	West	OFF-ST-10003306	Office Supplies	St
5157	CA-2017-163006	6/30/2017	7/4/2017	Second Class	GH-14410	Gary Hansen	Home Office	United States	Chicago	Illinois	60653	Central	TEC-PH-10002584	Technology	PH
6908	CA-2016-140249	9/27/2016	10/3/2016	Standard Class	SH-20455	Shaun Weien	Consumer	United States	Seattle	Washington	98103	West	TEC-PH-10002584	Technology	PH
2482	CA-2017-147956	12/24/2017	12/31/2017	Standard Class	AH-10210	Alan Hwang	Consumer	United States	Seattle	Washington	98105	West	OFF-ST-10001097	Office Supplies	St
409	CA-2017-117457	12/8/2017	12/12/2017	Standard Class	KH-16510	Keith Herrera	Consumer	United States	San Francisco	California	94110	West	FUR-TA-10002041	Furniture	Ta
1506	CA-2015-153388	8/1/2015	8/7/2015	Standard Class	PC-19000	Pauline Chand	Home Office	United States	Los Angeles	California	90004	West	FUR-TA-10003715	Furniture	Ta
1631	CA-2016-106530	5/8/2016	5/8/2016	Same Day	CL-12565	Clay Ludtke	Consumer	United States	Cleveland	Ohio	44105	East	OFF-ST-10002011	Office Supplies	St
949	US-2017-110576	11/28/2017	12/2/2017	Standard Class	RB-19795	Ross Baird	Home Office	United States	Philadelphia	Pennsylvania	19120	East	FUR-FU-10000576	Furniture	Fu
6579	CA-2014-103660	8/25/2014	8/30/2014	Standard Class	MW-18220	Mitch Webber	Consumer	United States	Seattle	Washington	98103	West	TEC-PH-10000895	Technology	PH
437	CA-2016-147375	6/12/2016	6/14/2016	Second Class	PO-19180	Philisse Overcash	Home Office	United States	Chicago	Illinois	60623	Central	TEC-MA-10002937	Technology	Ma
7082	CA-2016-106950	9/2/2016	9/6/2016	Standard Class	JE-15715	Joe Elijah	Consumer	United States	Charlotte	North Carolina	28205	South	TEC-AC-10002049	Technology	Ac
2877	US-2016-167339	1/21/2016	1/23/2016	Second Class	TD-20995	Tamara Dahlen	Consumer	United States	San Diego	California	92037	West	FUR-CH-10000785	Furniture	Ch
3769	CA-2014-153913	12/16/2014	12/20/2014	Second Class	KB-16585	Ken Black	Corporate	United States	Hialeah	Florida	33012	South	FUR-CH-10000988	Furniture	Ch
2240	CA-2016-146633	11/15/2016	11/17/2016	Second Class	TG-21310	Toby Gnade	Consumer	United States	Los Angeles	California	90049	West	OFF-BI-10003527	Office Supplies	Bi

products from Technology category

```
SELECT * FROM superstore
WHERE Category = 'Technology';
```

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category
27	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	TEC-AC-10003027	Technology
36	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	Gene Hale	Corporate	United States	Richardson	Texas	75080	Central	TEC-PH-10004977	Technology
41	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	Steve Nguyen	Home Office	United States	Houston	Texas	77041	Central	TEC-PH-10000486	Technology
42	CA-2017-120999	9/10/2017	9/15/2017	Standard Class	LC-16930	Linda Cazamas	Corporate	United States	Naperville	Illinois	60540	Central	TEC-PH-10004093	Technology
45	CA-2016-118255	3/11/2016	3/13/2016	First Class	ON-18715	Odella Nelson	Corporate	United States	Eagan	Minnesota	55122	Central	TEC-AC-10000171	Technology
48	CA-2016-169194	6/20/2016	6/25/2016	Standard Class	LH-16900	Lena Hernandez	Consumer	United States	Dover	Delaware	19901	East	TEC-AC-10002167	Technology
49	CA-2016-169194	6/20/2016	6/25/2016	Standard Class	LH-16900	Lena Hernandez	Consumer	United States	Dover	Delaware	19901	East	TEC-PH-10003988	Technology
55	CA-2016-105816	12/11/2016	12/17/2016	Standard Class	JM-15265	Janet Molinari	Corporate	United States	New York City	New York	10024	East	TEC-PH-10002447	Technology
60	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	Ted Butterfield	Consumer	United States	Troy	New York	12180	East	TEC-AC-10002167	Technology
63	CA-2015-135545	11/24/2015	11/30/2015	Standard Class	KM-16720	Kunst Miller	Consumer	United States	Los Angeles	California	90004	West	TEC-AC-10004633	Technology
69	CA-2014-106376	12/5/2014	12/10/2014	Standard Class	BS-11590	Brendan Sweed	Corporate	United States	Gilbert	Arizona	85234	West	TEC-PH-10002726	Technology

total sales by category

```
SELECT Category, SUM(Sales) AS `Total Sales`
FROM superstore
GROUP BY Category
ORDER BY `Total Sales` DESC;
```

Category	Total Sales
Technology	835900.0669999964
Furniture	733046.8612999996
Office Supplies	703502.9280000031

average profit by category

```
SELECT Category, AVG(Profit) AS `Average Profit`  
FROM superstore  
GROUP BY Category;
```

Result Grid		Filter Rows:
	Category	Average Profit
▶	Furniture	8.187449951783982
	Office Supplies	20.84239515654733
	Technology	79.05769255029902

total orders in each region

```
SELECT Region, COUNT(*) AS `Total Orders`  
FROM superstore  
GROUP BY Region;
```

Result Grid		Filter Rows:
	Region	Total Orders
▶	South	1579
	West	3099
	Central	2260
	East	2756

top 10 products based on total sales

```
SELECT `Product Name`, SUM(Sales) AS `Total Sales`  
FROM superstore  
GROUP BY `Product Name`  
ORDER BY `Total Sales` DESC  
LIMIT 10;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content
	Product Name	Total Sales		
▶	Canon imageCLASS 2200 Advanced Copier	61599.824		
	Fellowes PB500 Electric Punch Plastic Comb Bind...	27453.384		
	Cisco TelePresence System EX90 Videoconferen...	22638.48		
	HON 5400 Series Task Chairs for Big and Tall	21870.576		
	GBC DocuBind TL300 Electric Binding System	19823.479000000003		
	GBC Ibimaster 500 Manual ProClick Binding System	19024.5		
	Hewlett Packard LaserJet 3310 Copier	18839.686		
	HP Designjet T520 Inkjet Large Format Printer -...	18374.895		
	GBC DocuBind P400 Electric Binding System	17965.068		
	High Speed Automatic Electric Letter Opener	17030.311999999998		

top 3 categories based on sales

```
SELECT Category, SUM(Sales) AS `Total Sales`  
FROM superstore  
GROUP BY Category  
ORDER BY `Total Sales` DESC  
LIMIT 3;
```

	Category	Total Sales
▶	Technology	835900.06699999964
	Furniture	733046.8612999996
	Office Supplies	703502.9280000031

top 10 customers based on revenue generated

```
SELECT `Customer Name`, SUM(Sales) AS `Total Sales`  
FROM superstore  
GROUP BY `Customer Name`  
ORDER BY `Total Sales` DESC  
LIMIT 10;
```

Result Grid   Filter Rows:		
	Customer Name	Total Sales
▶	Sean Miller	25043.05
	Tamara Chand	19017.847999999998
	Raymond Buch	15117.339
	Tom Ashbrook	14595.62
	Adrian Barton	14355.610999999997
	Sanjit Chand	14142.333999999999
	Ken Lonsdale	14071.917
	Hunter Lopez	12873.297999999999
	Sanjit Engle	12209.438000000002
	Christopher Conant	12129.072

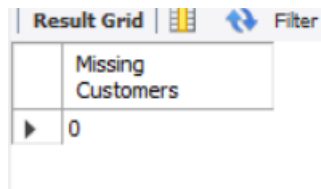
check duplicate order IDs

```
SELECT `Order ID`, COUNT(*) AS `Duplicate Count`  
FROM superstore  
GROUP BY `Order ID`  
HAVING COUNT(*) > 1;
```

Result Grid   Filter Rows:		
	Order ID	Duplicate Count
▶	CA-2016-152156	2
	US-2015-108966	2
	CA-2014-115812	7
	US-2015-118983	2
	CA-2014-143336	3
	CA-2016-137330	2
	CA-2016-121755	2
	US-2015-150630	6
	CA-2016-117590	2
	CA-2015-117415	4
	CA-2016-118255	2
	CA-2016-169194	2
	CA-2015-115742	4
	CA-2016-105816	2
	CA-2016-111682	7
	CA-2015-135545	4

check missing customer names

```
SELECT COUNT(*) AS `Missing Customers`  
FROM superstore  
WHERE `Customer Name` IS NULL;
```

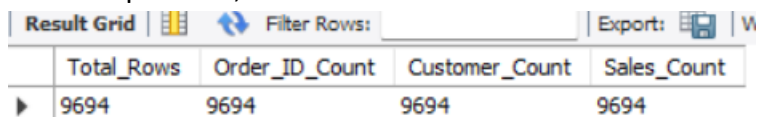


The screenshot shows a SQL query result grid with a single row. The column header is 'Missing Customers' and the value is 0. The interface includes a 'Result Grid' tab, a 'Filter' button, and a 'Filter Rows' dropdown.

Missing Customers
0

data quality check

```
SELECT  
COUNT(*) AS Total_Rows,  
COUNT(`Order ID`) AS Order_ID_Count,  
COUNT(`Customer Name`) AS Customer_Count,  
COUNT(Sales) AS Sales_Count  
FROM superstore;
```

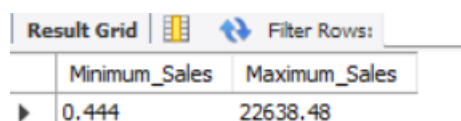


The screenshot shows a SQL query result grid with four columns: 'Total_Rows', 'Order_ID_Count', 'Customer_Count', and 'Sales_Count'. All four columns have a value of 9694. The interface includes a 'Result Grid' tab, a 'Filter Rows' dropdown, and an 'Export' button.

Total_Rows	Order_ID_Count	Customer_Count	Sales_Count
9694	9694	9694	9694

minimum and maximum sales values

```
SELECT  
MIN(Sales) AS Minimum_Sales,  
MAX(Sales) AS Maximum_Sales  
FROM superstore;
```



The screenshot shows a SQL query result grid with two columns: 'Minimum_Sales' and 'Maximum_Sales'. The values are 0.444 and 22638.48 respectively. The interface includes a 'Result Grid' tab, a 'Filter Rows' dropdown, and an 'Export' button.

Minimum_Sales	Maximum_Sales
0.444	22638.48

Insights from the Analysis

Dataset Overview

The Superstore dataset contains information about customer orders, products, sales, profit, regions, and order dates. Using SQL, the dataset was explored and analyzed to understand different aspects of the business data.

Sales Analysis

The sales data was analyzed using aggregate functions such as SUM() and AVG(). The queries helped identify the overall sales performance, compare different product categories, and determine the top-selling products based on total sales.

Customer Analysis

Customer-wise sales were analyzed to find the customers who contributed the highest revenue. This type of analysis helps in identifying valuable customers and understanding purchasing patterns.

Regional Analysis

The data was filtered based on different regions to observe the distribution of orders. The total number of orders was also calculated for each region using the GROUP BY clause.

Data Validation

Basic data validation checks were performed to verify the quality of the dataset. Missing values in important columns were checked, duplicate order IDs were identified, and row counts were compared to ensure the data was complete and consistent.

Conclusion

This assignment helped me understand how SQL can be used to analyze real-world datasets. I practiced filtering records, sorting data, grouping results, using aggregate functions, and validating data quality. Through this analysis, I gained practical experience in writing SQL queries to extract meaningful information from a dataset, which is an important skill for data analysis and database management.